

PRACTICAL NO. 5

AIM: In an operating system three CPU-intensive processes are ready for execution, which require 10ns, 20ns and 30ns and arrival at times 0ns, 2ns and 6ns, respectively. Write a program to calculate the total number of context switches needed if the operating system implements a shortest job first (preemptive) scheduling algorithm. Also calculate the average time for which the processes have to wait before getting the CPU.

CODE:

```
MINGW64/c/Users/airat/OneDrive/Desktop
GNU nano 8.7 sjf.c
#include <stdio.h>
#include <limits.h>

int main() {
    int n = 3;

    int process[] = {1, 2, 3};
    int arrival[] = {0, 2, 6};
    int burst[] = {10, 20, 30};

    int remaining[3];
    int waiting[3] = {0};
    int turnaround[3];
    int complete = 0, time = 0;
    int min = INT_MAX;
    int shortest = -1;
    int finish_time;
    int context_switch = 0;
    int prev_process = -1;

    float total_waiting = 0;

    // Copy burst time into remaining time
    for (int i = 0; i < n; i++)
        remaining[i] = burst[i];

    while (complete != n) {
        min = INT_MAX;
        shortest = -1;

        for (int j = 0; j < n; j++) {
            [ Read 80 lines ]
            ^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo     M-A Set Mark
            ^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/ Go To Line M-E Redo     M-6 Copy
```

OUTPUT:

```
Faiz@PUCKC3F09N9DT MINGW64 ~/OneDrive/Desktop (master)
$ nano sjf.c

Faiz@PUCKC3F09N9DT MINGW64 ~/OneDrive/Desktop (master)
$ gcc sjf.c -o sjf

Faiz@PUCKC3F09N9DT MINGW64 ~/OneDrive/Desktop (master)
$ ./sjf

Process Arrival Time    Burst Time    Waiting Time    Turnaround Time
P1              0             10             0              10
P2              2             20             8              28
P3              6             30            24              54

Average Waiting Time = 10.67 ns
Total Context Switches = 2

Faiz@PUCKC3F09N9DT MINGW64 ~/OneDrive/Desktop (master)
$ |
```

